BRK - Sonoscopeia

User manual and parameter reference · 1.0.1

See the notes, pitch glides and overlaps in your low end, or watch the full mix. Sonoscopeia draws the incoming sound without changing the audio.

Quick start

1. Insert Sonoscopeia after an instrument, drum bus or mix. Play audio through the track; no MIDI is required.
2. Set Range to the part of the spectrum you want to inspect. Start around the kick and bass, or Alt-click Range to return to 20 Hz–20 kHz.
3. Leave Sharp on for focused note and hit positions. Increase Resolution to separate nearby bass notes; use Freeze to inspect a moment.

Editor window

The editor opens at 1100 by 685 pixels and resizes down to 880 by 465. HELP opens a footer without changing the chosen width or moving the existing controls. Its Manual / PDF action opens this guide. Named controls support Tab, arrow keys and VoiceOver; the spectrogram picture and note readouts are visual. Press HELP again to close the help area.

Operation

Audio passes through unchanged in mono or stereo. Analysis runs in the editor on a background thread. Longer analysis windows delay the newest complete picture but add no audio-processing latency.

Sharp moves displayed energy toward its estimated time and frequency. It helps show a kick pitch drop or a bass glide clearly, but cannot separate tones closer than the chosen analysis window resolves. Clean removes much of the noise between tones, hits and glides; it only applies with Sharp enabled.

Resolution sets the analysis window for the low end. Longer windows reveal lower pitches and separate nearby notes; shorter windows follow changes sooner. The very lowest range is dimmed when the window does not contain enough cycles to place a frequency reliably.

Time chooses the history shown. Floor maps quiet energy to black and Ceiling sets the brightest level. A full-scale sine reads 0 dB; noise and hits do not have the same per-cell brightness calibration as sustained tones.

Mid displays the mean of left and right, Side their difference, and Left/Right inspect one channel. These options change only the analysis. Freeze holds the display while analysis continues; releasing it returns to the current picture.

Focused is the normal drawing mode. Pixels draws points at full strength, Original smooths the whole image and costs more drawing time, and Steady moves by a regular pixel pattern at the display refresh rate. Steady can use a

history span slightly different from Time; the displayed time labels show the actual span.

Controls

- **Resolution**: the analysis window in the lows, 21 ms to 1.4 s (170 ms to start with). Longer tells closer notes apart and sees lower; shorter follows faster changes. Above about 800 Hz it halves with each octave, down to 16 ms. The picture's right edge is half of it behind now.
- **Range**: the frequencies shown, any part of 2 Hz to 20 kHz at least a fifth wide (20 Hz to 20 kHz to start with), on a two-ended slider: drag an end, or between the ends to move both; a press past an end brings it there; Alt-click for 20 Hz to 20 kHz. Its ends are the Bottom and Top parameters. The picture so far is carried into the new range.
- **Time**: how much time the picture holds, 3 to 24 s.
- **Channel**: Mid (both channels together), Left, Right or Side (their difference).
- **Floor** and **Ceiling**: the levels drawn black and brightest. A full-scale sine reads 0 dB.
- **Sharp**: reassignment on. Off shows the classic spectrogram, for comparison.
- **Clean**: with Sharp, only tones, hits and glides.
- **Drawing**: Focused, Original, Pixels or Steady (above).
- **Colours**: the picture's colour scheme: Ember, Heat, Ice, Aurora, Phosphor or Mono.
- **Freeze**: holds the picture. The analysis goes on, so let go it carries on from now, unbroken, and the mouse still reads the held picture's frequencies. The audio still passes.
- **HELP**: opens a contextual footer for the hovered or keyboard-focused control. **Manual / PDF** opens the full reference at `https://barak007.github.io/brk-audio/manuals/sonoscopeia/`. Tab and VoiceOver reach named controls; the spectrogram and note readouts are visual. Help preserves the chosen width and leaves the controls in place.

Every parameter

Values and defaults are taken from the plugin. A control's effect can depend on the selected mode; see the operation notes above. Choices keep their saved identity between versions. Knobs and switches can be automated unless marked otherwise.

Parameter	What it does	Values and default
Resolution	Chooses the analysis window in the low end. Longer windows separate nearby bass notes and reveal lower pitches; shorter windows show changes sooner.	21 ms, 43 ms, 85 ms, 170 ms, 340 ms, 680 ms, 1.4 s Default: 170 ms Not host automatable
Bottom	Sets the lowest displayed frequency. Bottom and Top are the two ends of Range; the view enforces a minimum span.	2 Hz to 13.3 kHz Default: 20 Hz
Top	Sets the highest displayed frequency, limited by the input sample rate and the minimum span above Bottom.	3 Hz to 20.0 kHz Default: 20.0 kHz

Parameter	What it does	Values and default
Time	Chooses how much scrolling history the display holds. Steady drawing may use the nearest span that permits regular pixel movement.	3 s, 6 s, 12 s, 24 s Default: 6 s Not host automatable
Channel	Selects Mid, Left, Right or Side for analysis. The audio passed to the host is unchanged.	Mid, Left, Right, Side Default: Mid Not host automatable
Floor	Sets the quietest displayed level, mapped to black.	-140.0 dB to -40.0 dB Default: -96.0 dB
Ceiling	Sets the level mapped to the brightest colour, at least 1 dB above the effective floor.	-60.0 dB to +6.0 dB Default: 0.0 dB
Sharp	Enables reassignment to sharpen the displayed time and frequency positions. Off shows the classic spectrogram.	Off to On Default: On
Clean	With Sharp enabled, keeps coherent tones, hits and glides while reducing the noise between them.	Off to On Default: Off
Colours	Selects the display palette. This changes the picture only.	Ember, Heat, Ice, Aurora, Phosphor, Mono Default: Ember Not host automatable
Drawing	Chooses Original, Pixels, Focused or Steady drawing. See the operation notes for their appearance and drawing-cost differences.	Original, Pixels, Focused, Steady Default: Focused Not host automatable

Presets and files

The preset bar saves display and analysis settings. Init is the starting point; no factory preset bank is included. Save supports user folders and the preset-name menu manages saved presets.

User presets use .sonoscopeia files in ~/Library/Audio/Presets/Barakigal Audio/Sonoscopeia. The host session saves parameter settings and the selected preset. Freeze is a temporary editor action, not an audio or preset parameter.

Installation and help

Download the macOS archive from the product page and quit your audio host. Place the AU .component in ~/Library/Audio/Plug-Ins/Components, VST3 in ~/Library/Audio/Plug-Ins/VST3, or CLAP in ~/Library/Audio/Plug-Ins/CLAP. The product page also offers an installer with format selection. Restart or rescan the host. To uninstall, quit the host and remove that product's bundles from those folders; saved sessions and user presets are separate.

For questions or a problem report, use BRK Audio support on GitHub. Include the plugin version, macOS version, host and format, sample rate and a short description of how to reproduce it.